
AI-Driven Driver Redistribution

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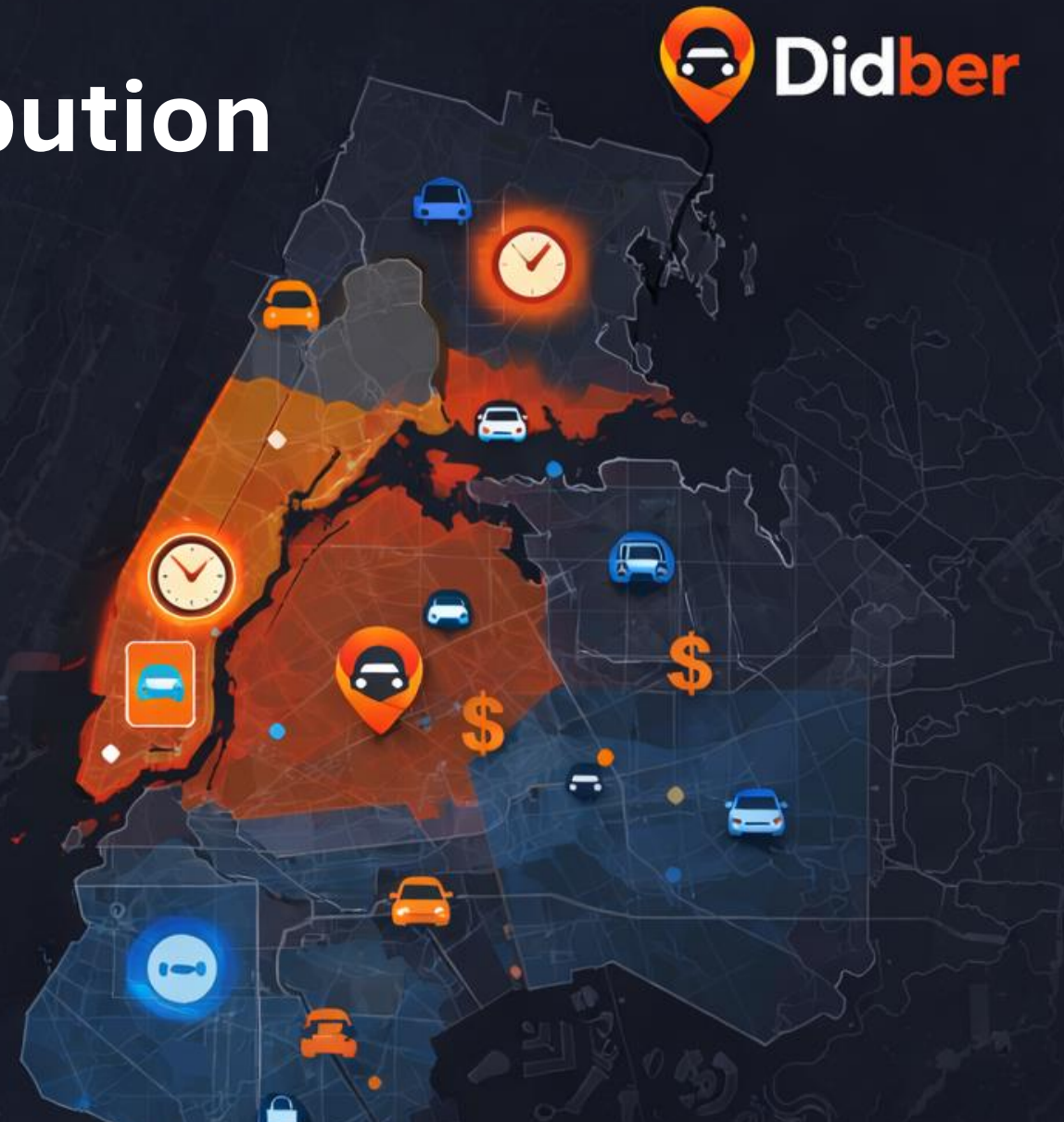
Assumptions and Context



- For this scenario, we assume we are working in the **Data Science team at Didber**, a ride-hailing and mobility platform like Uber, operating in large metropolitan areas.
- The analysis is based on **publicly available New York City TLC trip data from 2024 and 2025**, which includes detailed information on trip volumes, pickup and drop-off locations, timestamps, and demand patterns across different zones of the city. This dataset is used as a realistic proxy for Didber's operational data in a large, high-density urban market.

Inefficient Driver Distribution Limits Growth

- Driver supply is not aligned with real demand across zones and time slots.
- High-demand areas experience longer wait times and lost rides.
- Low-demand zones generate idle driver time and higher operational costs.



Executive takeaway:

Supply-demand imbalance is reducing revenue, customer satisfaction, and driver productivity.

AI-Based Driver Redistribution

Use AI to identify the optimal redistribution of drivers

Anticipate demand instead of reacting to it

Provide clear recommendations on where drivers should be positioned

Leverages existing operational data



Measurable Business Impact

- Reduced passenger wait times across high-demand zones
- Improved ride fulfillment during peak hours
- Better driver utilization and operational efficiency
- Direct improvement in customer experience and platform reliability

Executive takeaway:

Small optimizations in driver distribution translate into thousands of hours saved annually for passengers, directly impacting satisfaction and retention.



~27,500 passenger-hours saved per year

~750 passenger-hours saved per day

Generative AI for Real-Time Demand Signals (Next Step)

Leverage Generative AI to continuously scan public signals such as social media, event announcements, and local news

Detect upcoming events, large gatherings, or disruptions that may drive sudden increases in ride demand

Convert these early signals into actionable insights for proactive driver positioning

Generative AI allows Didber to move from forecasting demand to *anticipating it*, improving readiness, customer experience, and revenue capture.

